

IAN SEQUEIRA

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Research Interests: anytime-valid inference, e-value theory, compositional reliability for multi-agent LLM systems

EDUCATION

University of California, Santa Barbara

Santa Barbara, CA

B.S. Statistics & Data Science, B.A. Economics

Sept 2023 – June 2027

- GPA: 3.83 / 4.0 | Dean's Honors: 7 quarters | Graduate coursework: PSTAT 231 Statistical Machine Learning (A)
- **Selected coursework:** Design of Experiments (A+), Regression Analysis (A), Stochastic Processes, Data Science (A), Machine Learning, Econometrics (A), Probability & Mathematical Statistics, Proof Writing, Data Structures

RESEARCH EXPERIENCE

UCSB Department of Statistics & Applied Probability

Undergraduate Researcher (PSTAT 199) – Prof. Annie Qu (Founding Director, Center for Statistical Foundations of AI)

- Developing **Compositional Coordination Confidence (CCC)**, a framework for anytime-valid failure detection in multi-agent LLM pipelines using typed e-processes composed across DAG topologies
- Proved composition theorems establishing valid e-processes under sequential, fan-in, and fan-out coordination; Monte Carlo simulations confirm calibration and detection power across topologies
- Empirically validated on 1,642 annotated multi-agent traces across AutoGen, CrewAI, LangGraph, MetaGPT, and other frameworks, demonstrating early failure detection with controlled false-alarm rates
- Manuscript in preparation; ongoing work on adversarial coordination attacks and Bayesian priors over typed e-processes

Irvine Summer Institute for Biostatistics and Undergraduate Data Science (iSI-BUDS, NIH-funded)

Research Fellow – Prof. Dan Gillen & Prof. Josh Grill

- Constructed GLM, ridge, and CART classifiers in R to identify Alzheimer's biomarkers (pTau217, PACC, ADL) across 1,150+ A4 preclinical trial participants; best model ROC-AUC 0.835
- Developed biomarker-guided sample-size framework achieving 80% statistical power with 188 participants vs. standard 500+
- Contributing co-author on the resulting biomarker-guided trial design study; results appeared as a co-authored poster at *CTAD 2025*, San Diego (see Publications)

PUBLICATIONS AND PRESENTATIONS

Sequeira, I., Qu, A. *Compositional Coordination Confidence for multi-agent LLM systems*. Manuscript in preparation. 2026

Gaona-Partida, P., Anuurad, T., **Sequeira, I.**, Ragaine, E., Turner, C., Grill, J. D., Gillen, D. L. *Instructing Efficient Early-phase Preclinical Alzheimer's Disease Trials*. Manuscript in preparation. Preliminary results presented as a poster at *Clinical Trials on Alzheimer's Disease (CTAD)*, San Diego, CA, Dec 2025. 2025 – 2026

HONORS AND AWARDS

SIOP Summer Research Scholarship, UCSB Statistics & Applied Probability (mentor: Prof. Annie Qu) 2026

Hispanic Scholarship Fund (HSF) Scholar (renewable) 2024 – Present

Dean's Honors, UC Santa Barbara (7 quarters) 2023 – 2026

PROFESSIONAL EXPERIENCE

Alida Inc.

Remote

Data Engineering Intern

Sept 2024 – Present

- Architected a multi-mode AI data exploration platform on Databricks operating over 47 healthcare datasets, combining multi-agent routing, graph analysis, and automated SQL generation
- Extended the platform with a spec-driven ML prediction pipeline featuring typed contracts, dual-layer verification, and fault-tolerant LLM orchestration with AutoML training

Turing

Remote

Applied AI/GenAI Software Engineering Intern

Feb 2026 – May 2026

- Delivered full SDLC on a 13-node LangGraph call-triage pipeline (Intake Agent) during a 12-week industry sprint, sourced through the ACM Industry program at UCSB; work combined retrieval-augmented routing with typed agent contracts

TEACHING EXPERIENCE

Workshop Instructor, ACM at UCSB

2025 – 2026

- Led student workshops on applied machine learning and agentic LLM systems for undergraduate ACM members

SELECTED PROJECTS

Healthcare Financial Toxicity Prediction

PSTAT 231 (Graduate Statistical Machine Learning) final project

Mar 2026

- Predicted catastrophic out-of-pocket burden in 20,962 MEPS 2022 respondents; tuned 6 classifiers under 10-fold cross-validation, XGBoost reaching ROC-AUC 0.874
- Evaluated subgroup fairness across race/ethnicity using DALEX and `fairmodels`; interpreted feature contributions as SHAP odds ratios to surface actionable clinical predictors

Research Advisor Finder

Full-stack faculty discovery platform (FastAPI, PostgreSQL/pgvector, Next.js)

Nov 2025 – Feb 2026

- Matched student research interests to 15,000+ professors across 12 STEM fields via hybrid semantic search over CV-extracted interest embeddings

LEADERSHIP AND SERVICE

Project Executive, ACM Industry, UCSB

Sept 2025 – Present

- Leading cross-functional student teams delivering applied data science and software solutions for industry partners including Turing

Finance & Funding Chair, CampMed, UCSB

Sept 2025 – Present

- Leading budget planning and sponsorship strategy for health-equity outreach to underrepresented high school students

TECHNICAL SKILLS

Programming: R, Python, SQL, C++, TypeScript, LaTeX | **Methods:** e-processes, anytime-valid inference, GLMs, XGBoost, CART, SHAP, DALEX, Monte Carlo | **Tools:** tidymodels, rstan, scikit-learn, PyTorch, Databricks, LangGraph

PROFESSIONAL MEMBERSHIPS

American Statistical Association (ASA) | Institute of Mathematical Statistics (IMS) | UCSB Innovators Circle

REFERENCES

Prof. Dan Gillen, Chair, Department of Statistics, UC Irvine | **Prof. Annie Qu**, Founding Director, Center for Statistical Foundations of AI, UCSB | **Jay Chyung, MD, PhD**, COO, Alida Inc.
(contact information available on request)